



Short communication

Self-supported bimodal-pore structured nitrogen-doped carbon fiber aerogel as electrocatalyst for oxygen reduction reaction

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ABSTRACT

Self-supported 3-dimensional (3D) nitrogen-doped bimodal-pore structured carbon fiber aerogel is synthesized via a facile carbonization process using prawn shells as the raw material. The fabricated N-doped carbon fiber aerogel possesses micro- and meso-porous pores with an N doping level of 5.9% and a high surface area of 526 m² g⁻¹. As an electrocatalyst, the resultant N-doped carbon fiber aerogel exhibits superior electrocatalytic activity towards oxygen reduction reaction (ORR) with a more positive ORR onset-potential, better stability and high resistance to crossover effect compared to the commercial Pt/C electrocatalyst.

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1. Introduction

Recently, heteroatom-doped and co-doped carbon materials have been actively pursued as electrocatalysts for oxygen reduction reaction (ORR). Among them, N-doped carbon-based materials have been the most widely studied ORR electrocatalysts [1–3]. The reported theoretical studies indicate that the enhanced ORR activity of N-doped carbon materials can be attributed to the electron acceptability of doped N species that induces net positive charges to the adjacent carbon atoms to facilitate oxygen adsorption (the limiting step of ORR process) with low overpotential [1,4]. However, to obtain N-doped electrocatalysts, the synthetic procedures are either tedious or complex. More importantly, successful doping generally needs two or more precursors through either thermal treatment of pre-synthesized carbon or simultaneously formed carbon structure strategies [1,2]. Therefore, it is highly desirable to fabricate heteroatom-doped carbon materials by simple, low cost synthetic methods to replace expensive and scarce Pt-based ORR electrocatalysts. Biomass is a class of ideal starting materials for green synthesis of carbon-based ORR electrocatalysts because they are

earth-abundant, readily available and cheap to obtain [5]. Recently, our group reported the rational conversion of grass into N-doped carbon nanodot/nanosheet aggregates as high performance ORR electrocatalyst with comparable electrocatalytic activity, stability and resistance to crossover effects to the commercial Pt/C electrocatalyst [6]. Inspired by this, we intend to further develop a facile and effective method to produce carbon-based electrocatalyst from biomass such as plants or animal's shells in a large scale. As known, such materials have been treated as wastes that are normally disposed via burning and landfilling, inevitably causing the waste of valuable resources and pollutions to the environment. Chitin is a polysaccharide (molecular structure as shown in Fig. 1a, middle image) with very rich N content and the second most abundant biopolymer after cellulose, which has been widely used in a variety of areas such as biomedical, agriculture and water treatment applications [7]. The rich N content in chitin structure makes possible to directly transform it to N-doped carbon materials via a self-supported pyrolysis process without the need for additional N dopant source. It is well known that carbon-based aerogel possesses a unique 3D structure with low mass densities, open-pores and high surface area, advantageous for an ORR electrocatalyst. Therefore, it is highly desirable to realize the fabrication of N-doped, 3D carbon aerogel based high performance ORR electrocatalyst from biomass as sole precursor.

Herein, we use prawn shells as the raw material to firstly synthesize 3D structured chitin aerogel by a simple solution chemical route. After carbonization of the fabricated 3D chitin aerogel at 900 °C, the obtained N-doped carbon material maintains a 3D porous structure with high surface area, and exhibits superior ORR electrocatalytic activity.

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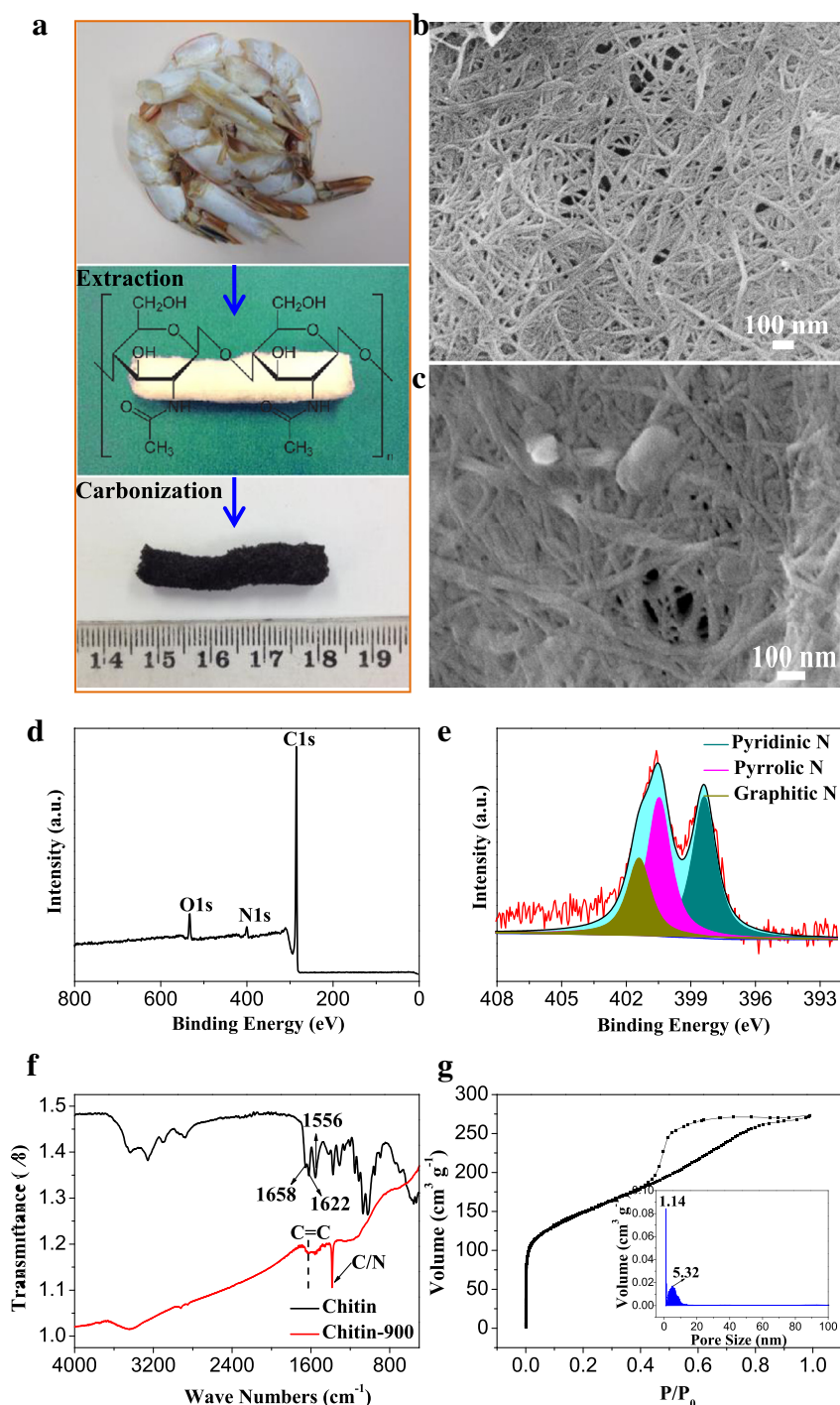


Fig. 1. (a) Fabrication procedure of the N-doped aerogel fiber from prawn shells. (b) and (c) SEM images of chitin aerogel before and after calcination at 900 °C under Ar protection, respectively. (d) XPS survey spectrum of Chitin-900. (e) Core-level N1s XPS of Chitin-900. (f) FT-IR spectrum of chitin and Chitin-900. (g) BET curve and pore size distribution (inset) of Chitin-900.

2. Experimental

2.1. Preparation of chitin aerogel and carbon-based electrocatalyst

Chitin nanofibers were prepared by previously reported method with minor modification [8,9]. Firstly, prawn shells were cleaned thoroughly, dried, milled to fine powders and then were refluxed in KOH to remove most of the proteins. Secondly, the powders were treated with HCl solution to remove the mineral salts. After filtration and rinsing with deionized water, the obtained powders were further treated in KOH solution to completely remove any residual proteins.

Thirdly, the pigment was removed from the powders using sodium chlorite solution. The 3D chitin aerogel was obtained by freeze-drying method. To achieve N-doped carbon-based material, the chitin aerogel was calcinated under the protection of Ar at 900 °C for 1 h with a ramp rate of 5 °C min⁻¹ (denoted as Chitin-900).

2.2. Material characterization

Scanning electron microscope (SEM) was employed for characterizing the morphology of the obtained sample. The chemical composition of the sample was analyzed by X-ray photoelectron spectroscopy

(XPS, Kratos Axis ULTRA incorporating a 165 mm hemispherical electron energy analyzer). Fourier transform infrared spectroscopy (FT-IR) analysis of the fiber was performed using Perkin Elmer spectrum 1000 FT-IR spectrophotometer with KBr as the reference sample. The Brunauer–Emmett–Teller (BET) method was utilized to calculate the specific surface areas (S_{BET}) using nitrogen adsorption–desorption isotherms of the sample on a Quantachrome Autosorb-1 equipment and the pore size distribution was fitted by the non-local density functional theory (NLDFT) [10].

2.3. Electrochemical characterization

Pre-treatment of the glassy carbon (GC) electrode can be found in our previously reported works [6,11]. To prepare the electrocatalyst suspension, 2.0 mg of Chitin-900 electrocatalyst was dispersed by sonication in 1000 μL of solvent mixture consisting of Nafion (5%), absolute ethanol, deionized water (V:V:V = 1:1:8). 12.0 μL suspension of the electrocatalyst was pipetted onto the GC electrode surface and allowed to dry at 60 $^{\circ}\text{C}$ for 10 min in a desiccator before use. For meaningful comparison, the Pt/C electrode with the same amount of the electrocatalyst was prepared using the same procedure as for the Chitin-900 electrode.

Electrochemical measurements were performed using a computer-controlled potentiostat (CHI 760D, CH Instrument, USA) with a typical three-electrode cell, incorporating a working GC rotating disk electrode (RDE, 5.0 mm in diameter) for the examination in the cyclic voltammograms (CVs) and linear sweep voltammograms (LSVs), an Ag/AgCl, KCl (3 M) electrode as reference electrode, and a Pt electrode as counter electrode. The CV experiments were conducted in O_2 -saturated 0.1 M KOH solution for ORR with a scan rate of 100 mV s^{-1} from -1.2 to $+0.2$ V at room temperature. LSV measurements were performed on RDE in the oxygen-saturated 0.1 M KOH solution at different rotation rates with a scan rate of 10 mV s^{-1} . The exact electron transfer number (n) was analyzed on the basis of Koutecky–Levich (K–L) equations shown in Eqs. (1)–(3) [6,11].

$$\frac{1}{J} = \frac{1}{J_L} + \frac{1}{J_K} = \frac{1}{B\omega^{1/2}} + \frac{1}{J_K} \quad (1)$$

$$B = 0.2nFC_0(D_0)^{2/3}\nu^{-1/6} \quad (2)$$

$$J_K = nFkC_0 \quad (3)$$

where J is the measured current density, J_K and J_L are the kinetic and diffusion limiting current densities, ω is the angular velocity of the disk ($\omega = 2\pi N$, N is the linear rotation speed), and n is the overall number of electrons transferred in oxygen reduction. According to Eqs. (1) and (2), the number n and J_K can be obtained from the slope and intercept of the Koutecky–Levich plots, respectively. F is the Faraday constant ($F = 96,485 \text{ C mol}^{-1}$), C_0 is the bulk concentration of O_2 ($C_0 = 1.2 \times 10^{-3} \text{ mol L}^{-1}$), ν is the kinematic viscosity of the electrolyte ($\nu = 0.1 \text{ m}^2 \text{ s}^{-1}$ in 0.1 M KOH), and k is the electron transfer rate constant. D_0 is the diffusion coefficient of O_2 in 0.1 M KOH ($D_0 = 1.9 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$). The constant 0.2 is adopted when the rotating speed is expressed in rpm.

3. Results and discussion

Fig. 1a shows the optical images of the prawn shell precursor, and the fabricated aerogel before and after pyrolysis. As can be seen, the shape of the obtained carbon aerogel is well preserved even after high temperature treatment. However, the color of the aerogel changed from white (middle photograph in Fig. 1a) to black (bottom photograph in Fig. 1a), indicating that the carbonization occurred during the thermolysis process. Moreover, as seen from Fig. 1b and c, the morphology of the obtained aerogel still shows a 3D porous fiber-like structure

after calcination, favorable for mass transport during electrocatalysis. Fig. 1d shows the XPS survey spectrum of the obtained carbon fiber aerogel sample (Chitin-900). The presence of N element confirms the successful self-supported doping after pyrolysis of chitin aerogel and an N doping level of 5.9% is achieved. The presence of O1s peak could be due to the physicochemically adsorbed oxygen, which might be an additional advantage for an ORR electrocatalyst [12]. The core-level XPS spectra of Chitin-900 sample (Fig. 1e) shows three types of N–C bonds centered at 398.3, 400.2, and 401.2 eV, corresponding to pyridinic-N, pyrrolic-N and graphitic-N, respectively [13]. The calculated percentages for pyridinic-N, pyrrolic-N and graphitic-N are 39.6%, 38.8% and 21.5% respectively. The above results can be further confirmed by FT-IR analysis (Fig. 1f). It is known that the FT-IR analysis is a conventional method to assess the purity of the chitin nanoparticles and to observe the presence of any residual surface bound proteins [14]. As seen from Fig. 1f, in the carbonyl region, the spectrum presents three strong peaks at 1658 and 1622 cm^{-1} , corresponding to the amide band I, and 1556 cm^{-1} corresponding to the amide band II are also observed. These strong peaks in the carbonyl region are special characteristics of the anhydrous α -chitin [15,16]. The absence of the peak at 1540 cm^{-1} confirms the absence of proteins in the obtained chitin [17], indicating that the successive treatments used in this work are sufficient to eliminate all the proteins to obtain pure chitin. Furthermore, the OH stretching band at 3482 cm^{-1} and NH stretching band at 3270 cm^{-1} of the chitin nanofibers are observed, indicating the presence of rich surface O- and N-containing function groups. After calcination, some characteristic bonds are observed and assigned to the stretching of carbonyl aromatic C=C bonds (1624 cm^{-1}). The bond at around 1383 cm^{-1} clearly seen in Chitin-900 sample can be attributed to the stretching mode of C and N bond heterocycles [18]. This evidence further indicates that N is successfully doped into the 3D aerogel structure. The broad band at around 3400 cm^{-1} in both curves can be attributed to the O–H or adsorbed H_2O molecules [6]. It is well known that large surface area combined with open porous structure is critically important for a high performance electrocatalyst as such structures facilitate the mass transport to allow the utmost utilization of all active areas of a material. Fig. 1g shows the N_2 adsorption–desorption isotherm for Chitin-900 sample. A high BET surface area of 526 $\text{m}^2 \text{ g}^{-1}$ is obtained. The curve shows a steeply increased adsorption at low relative pressures, indicating the presence of substantial micropore structures (<2 nm) [10,19]. The formation of hysteresis loops at relative pressures higher than 0.4 in the Chitin-900 sample indicates the existence of mesopores [20]. These results indicate that the formed catalyst possesses a bimodal-pore structure (pore size distribution centered at 1.14 and 5.32 nm corresponding to micro- and meso-pore, respectively), for which, the micro-pores provide a high surface area and the meso-pores facilitate the mass transport to allow the full utilization of the high surface area created by the micro-pores. Such advantageous structural characteristics would be highly beneficial to achieve high ORR electrocatalytic performance.

The resultant 3D N-doped carbon fiber aerogel is evaluated for its ORR electrocatalytic activity. The CV of Chitin-900 electrode in N_2 saturated KOH solution shows a typical pseudo-capacitive behavior of heteroatom-doped carbon, which can be found in Fig. 2a. A well-defined cathodic current peak at around -0.20 V resulting from O_2 reduction (OR) is observed from the O_2 saturated solution (Fig. 2a), indicating a pronounced ORR electrocatalytic activity. Theoretically, there should be only one cathodic peak centered around -0.2 V, resulting from a four-electron ORR process. The small cathodic peak observed at -0.35 V is likely due to the contaminations of electrode. However, it could be also due to the accompanied two-electron ORR process [21], although the four-electron transfer process is the predominant ORR process in this work.

To characterize the ORR process, LSVs of the electrocatalyst coated rotating disk electrode are carried out. As shown in Fig. 2b, the OR currents linearly increase with increasing rotation rates under the

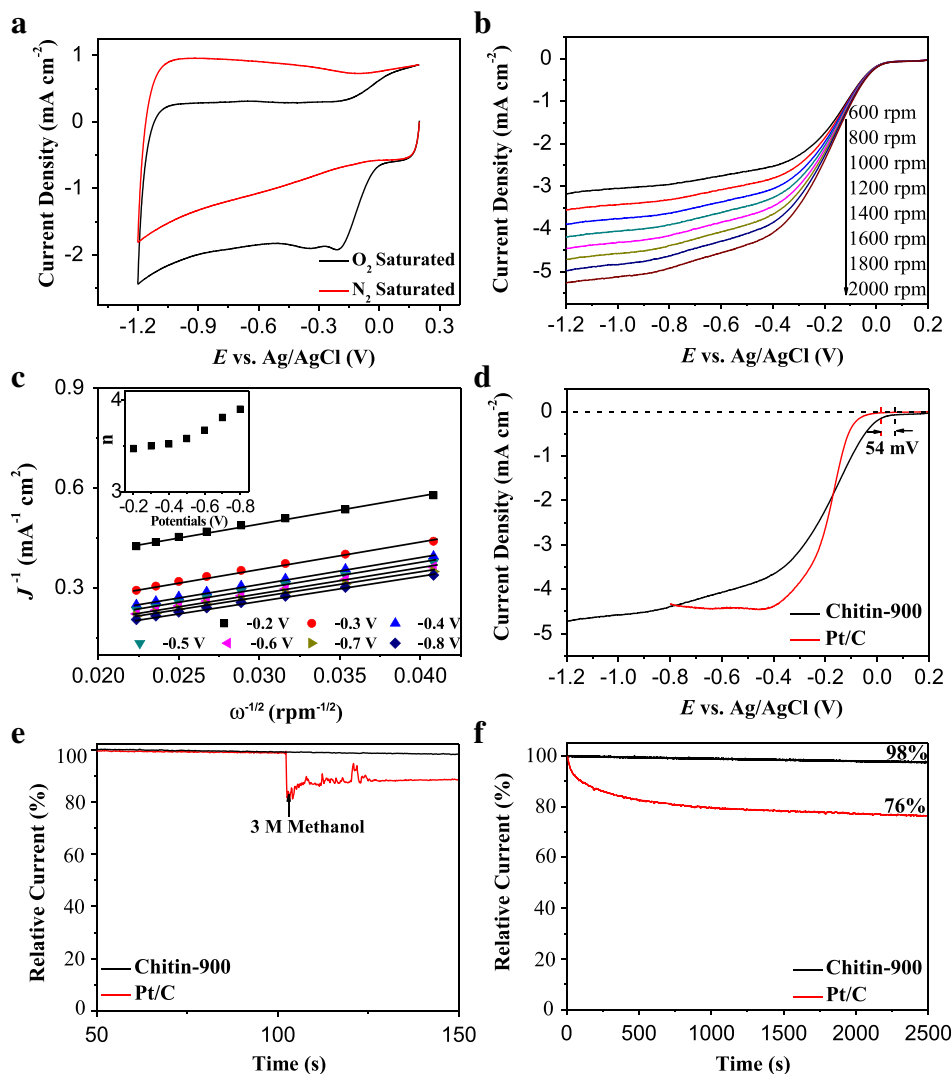


Fig. 2. (a) CVs and (b) LSVs of Chitin-900, respectively. (c) K–L plots and transferred electron number (inset) of Chitin-900 within the potential range from -0.2 to -0.8 V. (d) LSVs of Chitin-900 and Pt/C RDE electrodes at 1600 rpm and 10 mV s^{-1} . (e) Chronoamperometric responses on addition of 3 M methanol after about 100 s and (f) Durability evaluation at -0.4 V and a rotation rate of 1000 rpm for Pt/C and Chitin-900 electrodes, respectively.

conditions investigated. The corresponding K–L plots (Fig. 2c) of the electrode exhibit high linearity from -0.2 to -0.8 V. The transferred electron number (n) per oxygen molecule involved is calculated based on of the K–L equation. The transferred electron number (inset in Fig. 2c) increase from 3.47 to 3.90 within potentials ranged from -0.2 V to -0.8 V, signifying a typical four-electron ORR process. It is known that the onset-potential is an intrinsic thermodynamic character for an electrocatalyst and a less cathodic onset-potential indicates a lower ORR overpotential. For an ORR system, it is practically measured at the potential point where the cathodic current resulting from the oxygen reduction reaction begins [22]. As shown in Fig. 2d, Chitin-900 exhibits an onset-potential of 54 mV more positive than that of Pt/C catalyst (close to 0 V). Compared with other carbon-based electrocatalysts reported in the literatures, our result shows comparable or more positive onset-potential for ORR, indicating the superior electrocatalytic properties [1–4,23]. The excellent performance of the Chitin-900 electrode could be attributed to N-doping, especially the pyridinic-N which create large quantities of active sites (high charge and spin densities) that could play a key role in the ORR process [6]. Moreover, the high surface area and bimodal-pore structure of the aerogel also contribute to the excellent performance on the electrocatalytic behavior [23]. It is worth mentioning that for a kinetically controlled reaction, the rate constants determined by the K–L analysis

of RDE voltammetric data reflect apparent rate constants rather than the intrinsic ones in which the ratio of effective surface area of the electrode to the geometric area of the substrate (ψ) is not equal to 1 [24]. A consideration of the surface area ratio (ψ) should therefore be taken into account when interpreting the electrocatalytic effects using K–L analysis of RDE data. However, provided that the hydrodynamic requirements remain valid, the diffusion limited flux is independent of ψ and the determination of transferred electron number using the K–L equation is not affected [25].

It is well known that compared to Pt/C, heteroatom-doped carbon electrocatalysts possess high resistance to crossover effects by fuel molecules. Fig. 2e shows that Chitin-900 electrode can tolerate the presence of 3.0 M methanol without noticeable crossover effects, while a sharply decreased current is observed for commercial Pt/C electrode after the addition of methanol due to the methanol crossover oxidation reactions (MOR) [26]. It is known that the durability of an electrocatalyst for ORR applications is also critically important for practical use. As revealed in Fig. 2f, Chitin-900 and Pt/C electrodes are capable of retaining 98% and 76% of their initial activity after 2500 s of usage, respectively, indicating the Chitin-900 electrocatalyst synthesized in this work is more stable than that of Pt/C electrode. The superior durability of Chitin-900 electrode can be ascribed to the superior 3D structural stability.

4. Conclusion

In summary, 3D N-doped carbon fiber aerogel has been successfully fabricated using prawn shells as the raw material. The 3D aerogel possesses a bimodal-pore structure and high surface area. As an electrocatalyst, the N-doped carbon material shows a typical four-electron oxygen reduction process. Moreover, the obtained catalyst exhibits superior electrocatalytic activity, good stability and high resistance to the methanol crossover oxidation reaction. The findings of this work would be helpful for design and fabrication of high performance N-doped carbon-based electrocatalysts using biomass as the precursor.

Conflict of interest

The authors declare no conflict of interest.

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